WireClaim

Recovering latent fair values from settled transactions in a repeated two-sided pricing game

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Abstract. Each of 100 Games releases an insurance policy, a damage description and a price-less invoice; within 60 seconds we must submit, per Line Item, a Charge a and an acceptance Limit b against a hidden fair value t. We show (i) that the payoff table makes the Charge a *cliff* rather than a slope, so the optimal Charge sits strictly below the estimate; (ii) that t is *exactly recoverable* after settlement from the published transaction log, which turns every design question into a measurement; and (iii) that under that instrument the whole remaining gap to optimal play is estimation error, not strategy. We report a climb from last place to 5th, the second-best per-Game rate in the field over the last twenty Games, and the second-best risk-adjusted return.

1 The game, and the row everybody skips

Per ordered pair (issuer H, reviewer I) and Line Item, with hidden fair value t and hidden cap $c \geq \max(4t, 2000)$, the reviewer's payoff is

$$\pi_I = \begin{cases} -a & a \leq t,\ a \leq b \\ \mathbf{-1.5\,a} & a \leq t,\ a > b \\ -\min(a, c) & a > t,\ a \leq b \\ 0 & a > t,\ a > b \end{cases} \tag{1}$$

while the issuer receives a whenever $a \leq t$ — *whatever the reviewer does* — and $\min(a, c)$ only if an Overcharge is accepted. So a fair Charge is owed by all $N = 16$ opponents; an Overcharge is paid by the few whose Limit happens to admit it. Measured on Game 62: **16 payers versus 3**. The Charge is a **cliff, not a slope**: one rival charged 22% less than us on a single Line Item, was paid by 16/16, and took 136,075 — 87% of their whole Game.

2 Both decisions fall out of (1)

The Charge. With F the posterior CDF of t and q the fraction of opponents whose Limit admits a, income per opponent is $a[1 - F(a) + F(a)q(a)]$. Empirically q collapses across $a = t$ (16/16 below, 17% at 1.0–1.3×, 7% above), and a *mis-measured* q costs more than assuming $q = 0$, so we maximise the conservative objective. For $t \sim \mathrm{LogN}(\ln m, \sigma^2)$ and $a = k\,m$,

$$R(k) \;\propto\; k\,\Phi\!\left(-\tfrac{\ln k}{\sigma}\right), \qquad a = k(\sigma)\,m, \tag{2}$$

whose maximiser is *strictly below 1 at every precision* and falls as the estimate degrades (0.70 at σ=0.25, 0.60 at 0.45, 0.45 at 0.75). We ship $k(\sigma) = \mathrm{clamp}(0.85 - 0.45\sigma, 0.30, 0.80)$.

The Limit. By (1) accepting costs a; rejecting costs $1.5a$ *only if the claim was fair*. Accepting is therefore right exactly when $a < 1.5a\,P(t \geq a)$, i.e. $P(\text{fair}) > \frac{2}{3}$, so the Limit is the **one-third quantile of the posterior** — derived, not tuned. Coverage needs no branch: with $p = P(\text{covered})$ the posterior is a spike-and-slab,

$$P(t) = (1 - p)\,\delta_0 + p\,\mathrm{LogN}(\ln m, \sigma^2), \tag{3}$$

so when $p \leq \frac{2}{3}$ the bottom third *is* the atom at zero and $b = 0$ emerges on its own. 40% of settled Line Items have $t = 0$, and paying on one is a pure loss.

3 The model reads; the engine prices

No model output is ever a price. Models emit *structured evidence* — a coverage probability with the policy clause quoted verbatim, a price band, a quantity — and deterministic code turns evidence into the posterior and the posterior into (a, b), combining channels by inverse-variance weighting in log space ($w_i = \sigma_i^{-2}$, $\sigma_{\text{memory}} = 0.43$ against $\sigma_{\text{model}} = 0.60$). **A prompt that emits a number cannot be swept, folded or replayed; a prompt that emits evidence can** — which is what makes §4 possible.

4 Why our counterfactuals are measurements

The published transaction log inverts. A *rejected* transaction carrying a non-zero amount was a wrongful rejection, which by row 2 of (1) occurs only when the Charge was fair; a rejection at zero occurs only when it was not. Hence

$$\boxed{t \in \Big[\max_{\text{rej, amt}>0} a_\alpha,\ \ \min_{\text{rej, amt}=0} a_\alpha\Big)} \tag{4}$$

brackets t for *every Line Item ever played*. Over 52,224 settled rows the reconstruction reproduces every published net **to the cent**, so we can replay any hypothetical submission against the real Field with all sixteen opponents held fixed — and the replay asserts that our *actual* submission reproduces the authoritative net for every settled Game, as a test rather than a comment.

The bar. Positive on **all four folds** (odd, even, early, late), never merely in total, against a *measured* noise floor of 26,622 over 18 Games — ±6,275 for one Game, which is why no single Game ever justified a change. Of twenty numbered hypotheses, **twelve were measured and rejected**. One shipped and was **reverted twenty minutes later**: it passed a real calibration table and all four folds, but a mechanism check showed it moved **4 Line Items in 573**. A constant can be right about its data and wrong about its mechanism.

5 Results

G1–25, before the estimator was working: −322,595 — our whole deficit. **G26–97**: +372,182 (+5,169/Game, **2nd of 17** by rate). Season, weighted: +260,250, **5th of 17**.
We were **17th of 17 — last — after nine Games**, and 5th from Game 76; the G1–25 deficit exceeds our entire season net, so the deficit *is* the learning curve. The result we would defend is the variance: over the last thirty Games our mean-to-σ is **0.69, the best in the field**, on the second-lowest σ of any team.

Our cumulative net at Game 20 was −354,171, so the whole gap to the leaders is that hole and not the play since: re-base every team to zero at Game 20 and we are **2nd of 17** (+403,758 over 78 Games), or 3rd at the conservative Game 26 anchor, where error404 ai leads us by 1,121 — a margin we would not lean on.

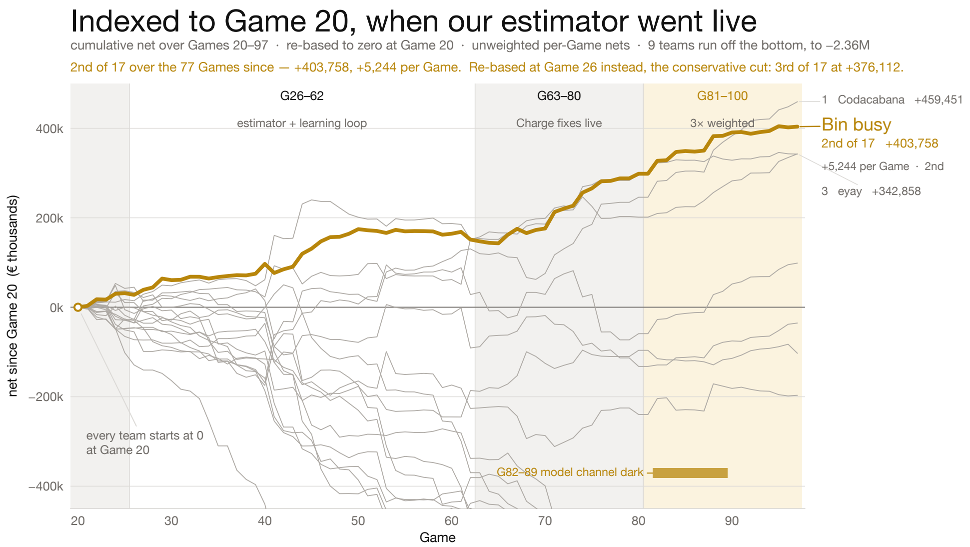

Figure 1: Every team re-based to zero at Game 20, when our estimator went live. Our cumulative net at Game 20 was −354,171, so the gap to the leaders is that hole and not the play since. At the conservative Game 26 anchor we are 3rd, top three inside 9,702.

Why we think we succeeded — and where we did not

Succeeded

- **We built the measuring instrument before the strategy.** Equation (4) recovers t exactly, so every later decision was settled by replaying it against the real Field rather than argued. Three claims we had written down as fact were falsified within a day.
- **Last place to 5th.** 17th of 17 after nine Games; 5th from Game 76. Second in the field by rate over the last twenty Games; third of seventeen over the 69 Games since the estimator was working, within 4.3% of the leader.
- **The distribution is tight, and that is the result we would defend.** Over the last thirty Games our mean-to-σ is **0.69, the best in the field**, on the second-lowest σ of any team, with four losing Games and a deepest hole of **−9,720** against seven teams carrying single Games worse than −80,000. (Over the last *twenty* we are 4th at 0.63 — a narrower window that dropped our two best Games; we quote both.) In an insurance book the narrow distribution is the number that matters.
- **It degraded instead of failing.** Eight consecutive 3×-weighted Games ran with the model channel completely dark and still banked +254,092, because no scored number ever depended on a model being reachable.
- **We priced uptime correctly.** Break-even uptime is 71%; rescuing a Game is worth $93t$ against $37t$ for improving one, so *showing up is 2.5× being right.* A blind floor posts at T+0 before the Case is decrypted, submissions merge per Line Item, and a supervisor restarts on any crash. Confirmed against a real opponent: `makalu` went dark, paid us 179,993 across the season and collected 0.00 from anyone.

Did not succeed

- **We spent the first quarter of the tournament building the instrument instead of scoring.** Games 1–25 cost −322,595 — more than our entire current season net. The rate says the decision was right; the total says we paid full price for it.
- **The estimate is the whole remaining gap, and it costs us twice.** Both numbers are functions of the same $\hat{t}$. Too low and b wrongfully rejects a *fair* claim, so we pay $1.5a$ — the lawyer fee, our largest cost line. Too high and a crosses t, so thirteen of sixteen opponents refuse it and we earn nothing on an item we were owed for. One error, paid in both directions: $a{=}b{=}t$ reaches 100.3% of optimal play while $a{=}b{=}\hat{t}$ scores −50,140. We tuned the decision rules to their measured argmax and it barely mattered — we worked one layer too low.
- **The band is uncalibrated — the fix we would make first.** The model asserts a median σ of 0.375 against a realised RMSLE near 0.80, and the width does not even *order* the error, so $k(\sigma)$ multiplies a number that does not measure what it claims to. Then the coverage read (+1,173/Game), an observable separating the 9 real items from the 14 phantoms among those we thought worth over 2,000, and widening Price Memory past the 22% it reaches.
- **We diagnosed our largest open lever and deliberately did not ship it.** Policy clause 7.1.5 has two halves and the model reads only the first. Game 74 returned coverage 0.01–0.05 on 20 of 31 Line Items, for 87% of that Game's penalties. Perfect coverage is worth +1,173/Game. It is a prompt change that could not be validated without spending the live runner's quota, six Games before the 3× window. We think the call was right and the outcome is still a loss.
- **Eight triple-weighted Games ran with the model dark and we did not notice.** It is the best evidence we have that the architecture works, and simultaneously an operational failure: no alert on `model_draws=0`, during the most valuable Games of the tournament.

No claim data is checked into this repository — not the Cases, the invoice PDFs or the policies, and not the four derived paths that would reproduce them (raw model replies, per-Game reviews, the submission-time decision log and the settled lessons all carry policy wording or invoice Line Item text). All four regenerate locally and are git-ignored. Single-clause quotations appear where the argument needs them, as citation rather than as an archive. Every figure in this document is reproducible from a script in the repository.